Little Explorers · Week 5

Automatic Night Lamp

A lamp that knows when it is dark!

Ages 5-7 | Arduino + light sensor | 60 minutes



Did you know?
Many streetlights switch on by themselves every evening. Nobody presses a button!

Let's think...

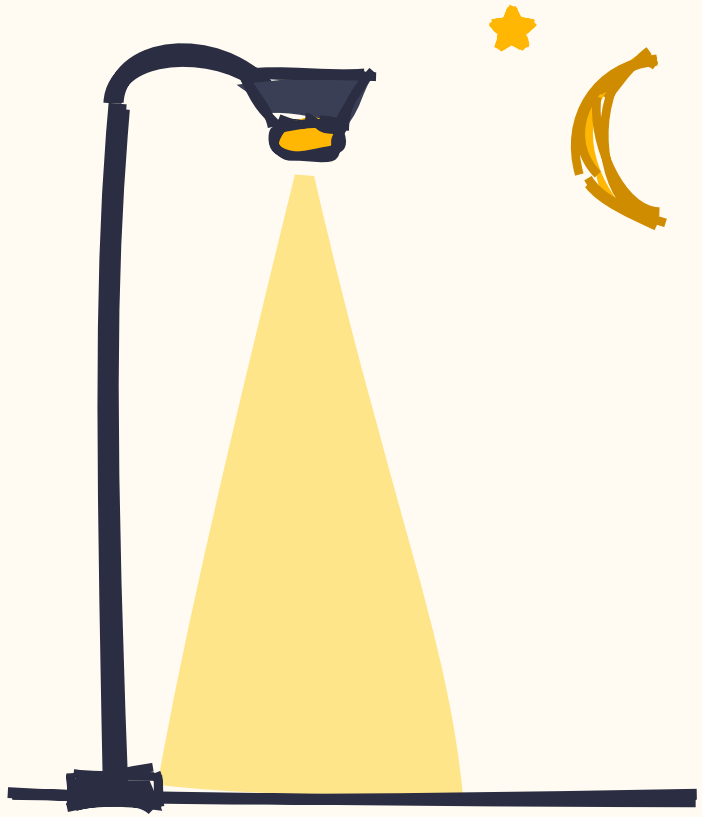
Who switches on the streetlights?

Every evening, thousands of streetlights turn on.
Does someone climb every pole to switch them on?

A helper with a ladder?

A magic button?

The lamp itself?

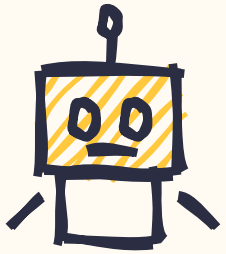


Big secret
The lamp has a tiny eye that feels the light. Today we will build one!

LEARN

Our learning aim

Today, I will learn to...



1 I can tell that **light** and **dark** can be measured.

2 I can use a **light sensor**, the lamp's eye.

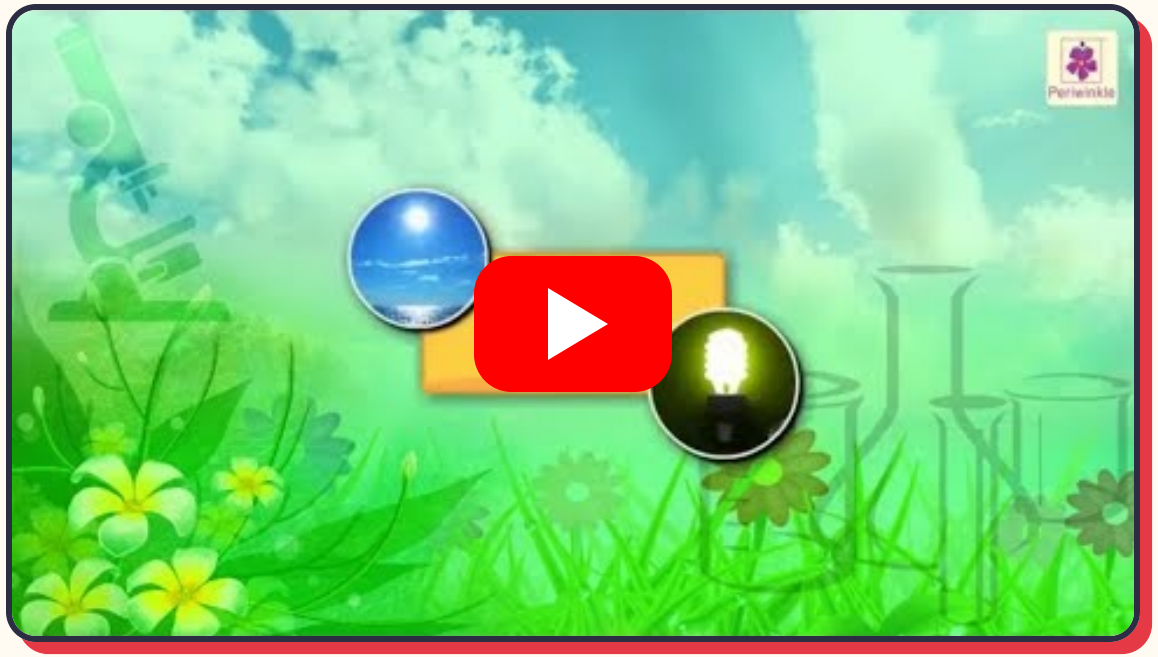
3 I can use an **if-then** block to switch on a light in the dark.

Keywords

light · dark · sensor · if-then · automatic

Watch and think

Video time!



Light

Periwinkle · about 2 minutes · youtu.be/eAK54FA0vWI

Watch for...

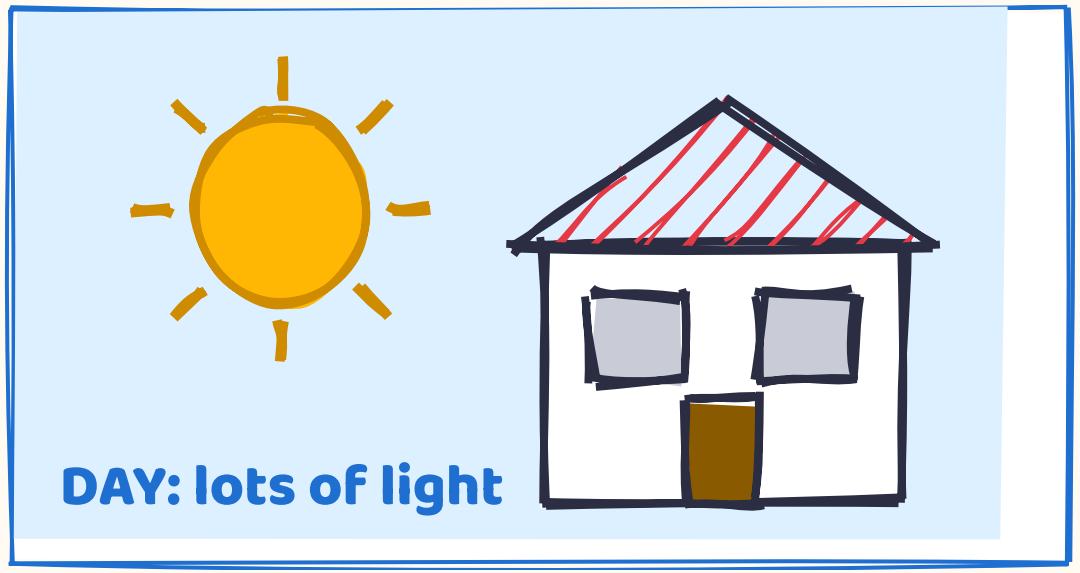
1. Where does **light** come from?
2. Name one thing that **makes** light.
3. What happens when it gets **dark**?

Scan to watch

After the video: Our lamp turns on when it is dark. Cover the sensor with your hand: what will happen?

Look around you

Day and night



Lamps can stay **OFF**



Lamps must turn **ON**

Try it: cover your eyes with your hands. Dark! Now open them. Light! Your eyes are **light sensors**.

LEARN

Meet our new friend

The light sensor: the lamp's eye



1. Sensor feels light
2. Arduino decides
3. Lamp turns on

What is a sensor?

A sensor helps a machine **feel** the world, like our eyes, ears and skin help us.

Its real name

This sensor is called an LDR: a Light Dependent Resistor. Grown-up word!

Sensor (input) → brain (decision) → lamp (output). Every robot works like this!

Look around you

Light sensors are everywhere!



Garden solar lights
Charge in the sun, glow at night



Car headlights
Some cars switch lights on in tunnels



Phones and tablets
Screen gets brighter in sunshine



Bedroom night lights
Glow softly when you turn off the big light

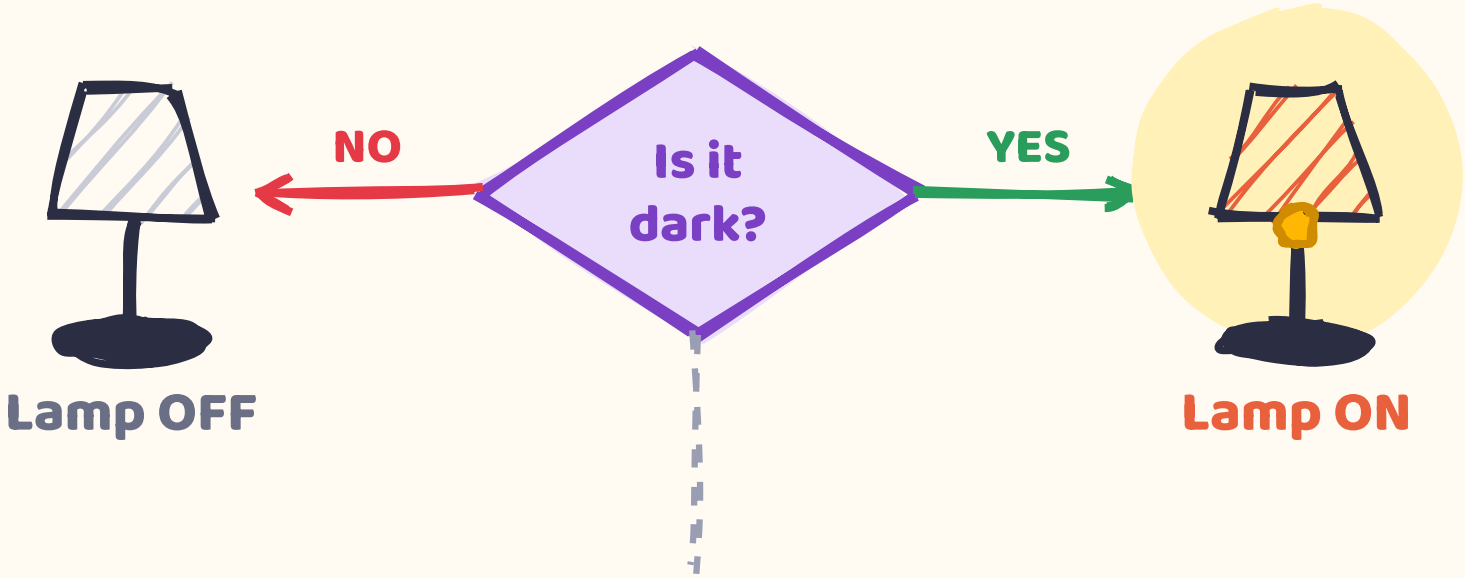
Saving energy: lights that switch off in the daytime do not waste electricity. That is good for our planet!

Spot it!
How many automatic lights can you find at school or at home?

LEARN

How the lamp thinks

IF it is dark, THEN turn on



Robot rule

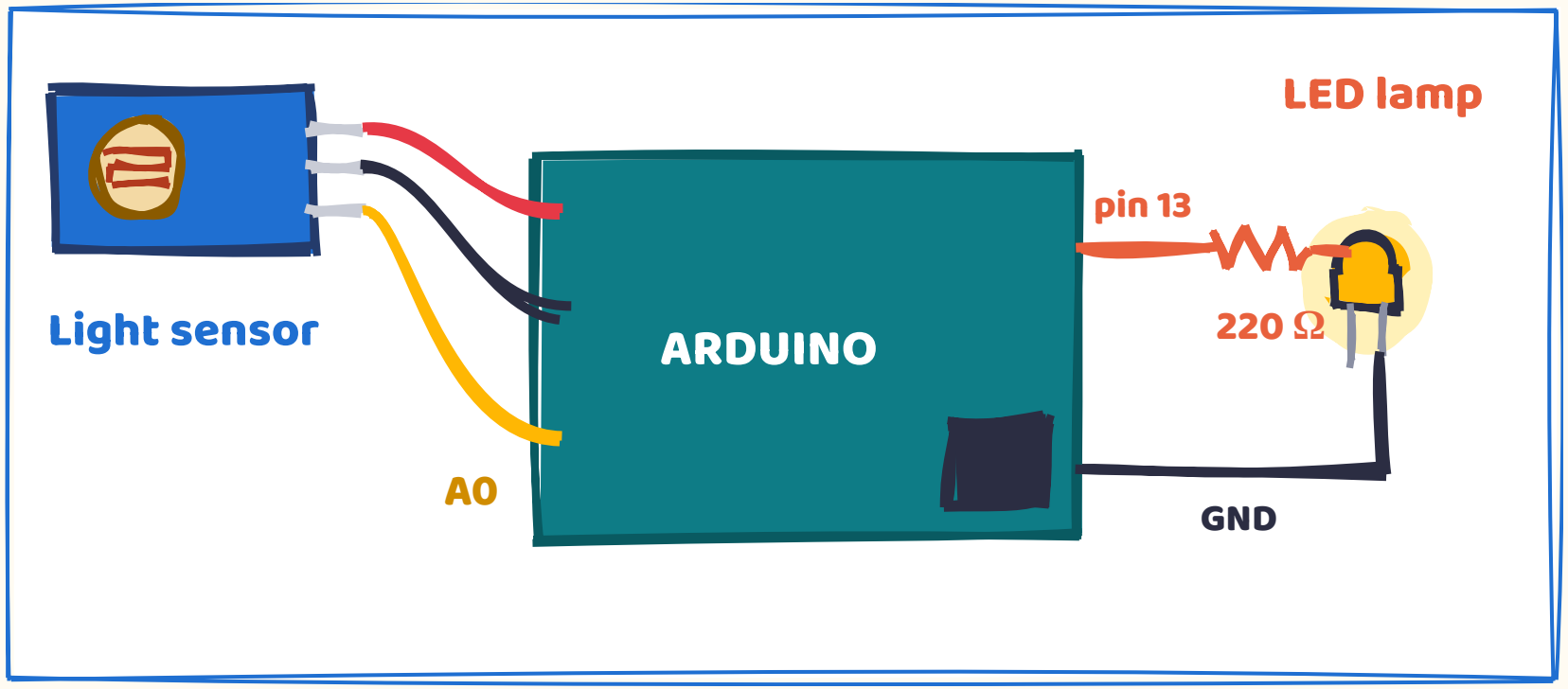
IF something is true,
THEN do this.
ELSE (if not), do that.

You use rules too

IF it is raining, THEN I take my umbrella!

Look inside

How our lamp is connected



Colour clues

Red: power (5V)

Black: back home (GND)

Yellow: sensor message to A0

Orange: LED on pin 13

Teacher: the circuit is pre-wired. Children plug in the USB cable, test and program.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Be the night lamp!
2. Meet our tools
3. Find your light number
4. Let's code our night lamp
5. Does our lamp work?

Stand up and play!

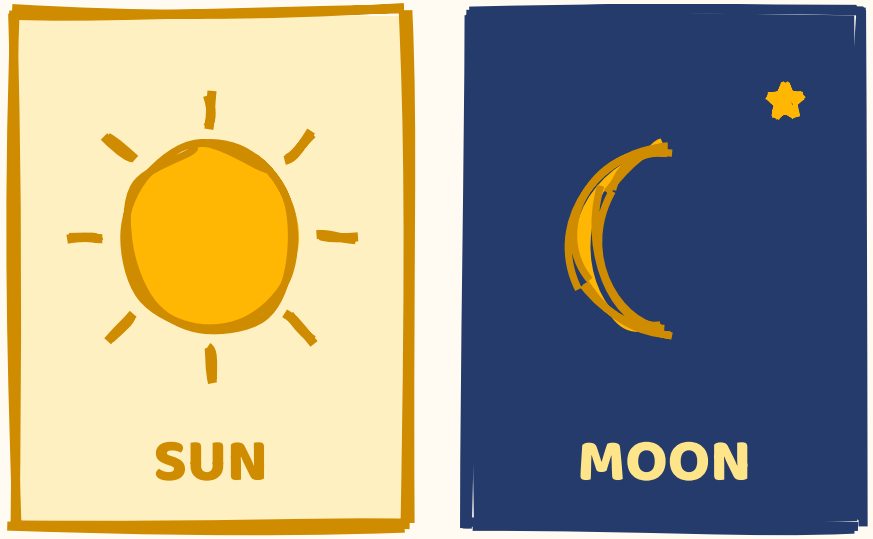
Be the night lamp!

How to play

Your teacher is the **light sensor**.

Sun card: sit down, lamp OFF

Moon card: stand up and shine like a star, lamp ON!



Listen carefully: **IF** it is dark, **THEN** the lamp turns on. That is a rule!

Tricky round
Half sun, half moon? Let's decide together what the lamp should do!

MAKE

Our builder's kit

Meet our tools

Arduino
The robot brain

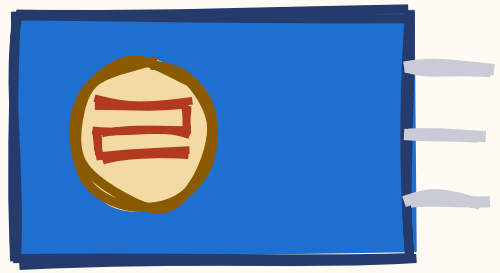
Light sensor
The lamp's eye

LED
Our tiny lamp

Wires
Roads for electricity

Laptop
Where we snap our blocks

Torch
Our pretend sunshine



light sensor

In real life
The same kind of tiny computer is inside washing machines, toys and real streetlights!

MAKE

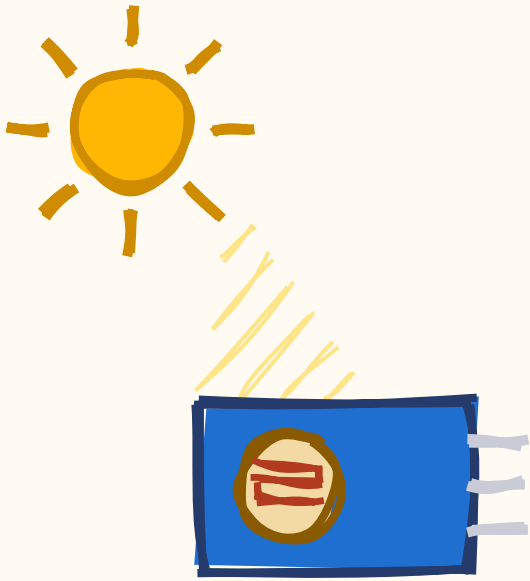
Be a scientist!

Find your light number

The sensor gives us a number

1. Shine the **torch** on the sensor. Write the number.
2. **Cover** it with your hand. Write the number.
3. Pick a number **in the middle**. That is where day turns into night!

Torch ON (bright)	my number: _____
Hand cover (dark)	my number: _____
Middle number	my number: _____



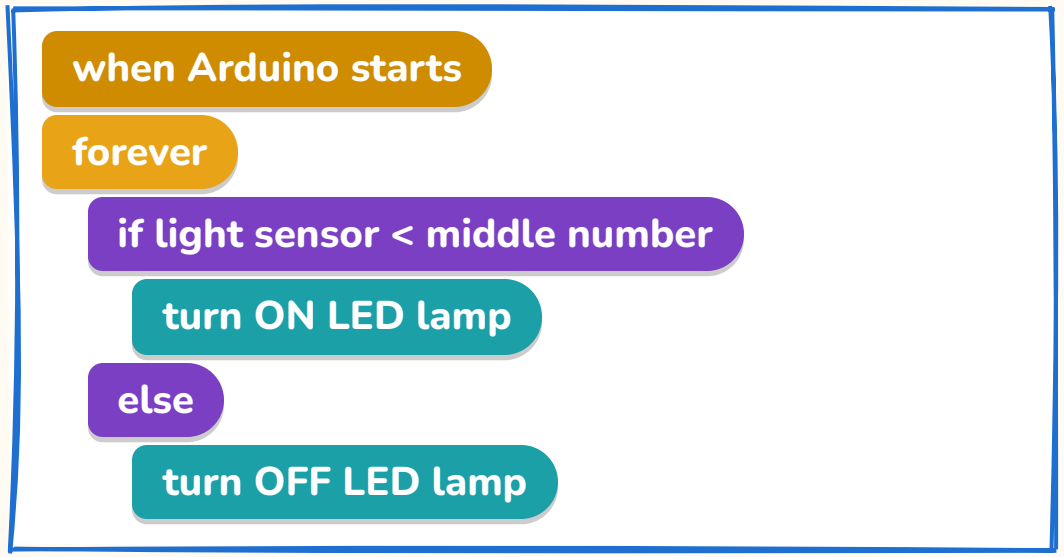
light sensor

Measure it!
Light is something we can count with numbers.

MAKE

Time to build!

Let's code our night lamp



3 easy steps

1. Snap the blocks
2. Type your **middle number**
3. Upload and cover the sensor!



Magic words: IF and ELSE
IF it is dark, the lamp turns ON. ELSE it turns OFF.

Teacher: some sensor modules count the other way round. If the lamp lights in the bright, swap < for >.

Test and improve

Does our lamp work?



Shine the torch. Lamp ON or OFF? Guess first, then check!



Cover with your hand. Lamp ON or OFF? Guess first, then check!



Switch off the class lights. What happens? Guess first, then check!

Challenge time!



Sunset lamp: add an orange LED for evening, when it is only a little dark.



Goodnight beep: add a beep when the lamp turns on.



Bug hunt: the lamp is on in the day! Can you fix it?

Engineers test, test, test!
Real streetlights are tested in bright days, dark nights and cloudy weather.

Show what you know

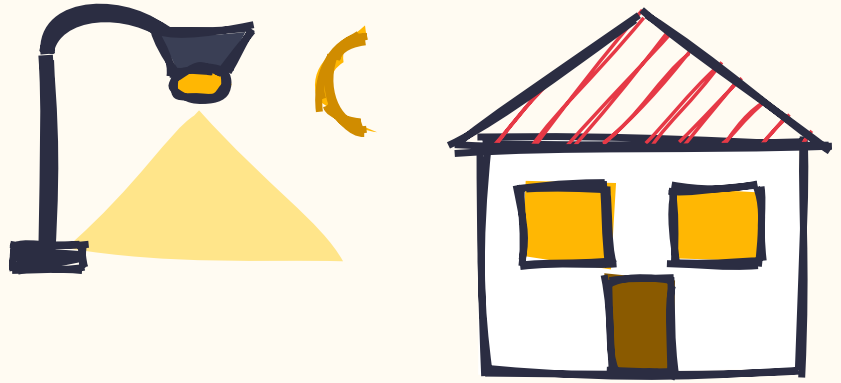
What did we learn?

- [x] Light and dark can be measured with numbers
- [x] A light sensor is the lamp's eye
- [x] IF it is dark, THEN the lamp turns ON
- [x] Automatic lights save energy

Exit ticket: thumbs up if you can tell a friend how the lamp knows it is dark!

Tell your family tonight

Watch a streetlight at sunset. Does it switch on by itself? Tell us next week!



Next week: a Hot & Cold Temperature Indicator!